

Qiskit v2.x Practice Quiz

Questions

1. Which code correctly creates a two-qubit quantum circuit?
 - A. `QuantumCircuit()`
 - B. `QuantumCircuit(2)`
 - C. `Circuit(2)`
 - D. `QuantumRegister.create(2)`
2. What does layout refer to in transpilation?
 - A. Mapping virtual qubits to physical qubits
 - B. Choosing the number of shots
 - C. Drawing the circuit
 - D. Choosing classical register names
3. Which object is commonly used to represent Pauli observables in Qiskit?
 - A. `QuantumCircuit`
 - B. `SparsePauliOp`
 - C. `ClassicalRegister`
 - D. `PassManager`
4. Which gate maps $|0\rangle$ to $|+\rangle$?
 - A. X
 - B. Z
 - C. H
 - D. CX
5. When is Sampler more appropriate than Estimator?
 - A. When you need bitstring measurement outcomes
 - B. When you need expectation values of observables
 - C. When you need to draw a circuit
 - D. When you need to create a backend
6. What does increasing the number of shots usually improve?
 - A. Sampling precision

- B. Circuit depth
 - C. Qubit connectivity
 - D. Gate fidelity automatically
7. OpenQASM is best described as:
- A. A language for representing quantum programs
 - B. A physical quantum processor
 - C. A Python plotting library
 - D. A classical database system
8. Which visualization is commonly used for measurement counts?
- A. Histogram
 - B. Bloch sphere only
 - C. Circuit drawer only
 - D. Density matrix table only
9. Why may SWAP gates be inserted during transpilation?
- A. To reset qubits
 - B. To satisfy hardware connectivity constraints
 - C. To increase shot count
 - D. To remove all noise
10. What does the Estimator primitive primarily compute?
- A. Bitstring counts
 - B. Expectation values of observables
 - C. Circuit diagrams
 - D. Backend coupling maps
11. What is OpenQASM primarily used for?
- A. Describing quantum circuits
 - B. Training classical neural networks
 - C. Managing cloud billing
 - D. Creating PDF files
12. What does the Bloch sphere represent?
- A. Only classical bits
 - B. Single-qubit pure states
 - C. Only two-qubit entangled states
 - D. Classical probability distributions only
13. What does the Sampler primitive primarily return?

- A. Expectation values
- B. Measurement samples or bitstring results
- C. Only statevectors
- D. Only transpiled circuits

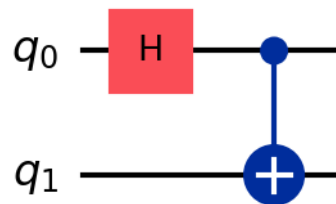
14. What is the result of applying Z to $|1\rangle$?

- A. $|0\rangle$
- B. $|1\rangle$
- C. $-|1\rangle$
- D. $|+\rangle$

15. Which state lies on the positive X axis of the Bloch sphere?

- A. $|0\rangle$
- B. $|1\rangle$
- C. $|+\rangle$
- D. $|-\rangle$

16. What state does the circuit below prepare?



- A. $|00\rangle$
- B. $|11\rangle$
- C. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- D. $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

17. What does the following circuit prepare?

```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
```

- A. $|00\rangle$
- B. $|11\rangle$
- C. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- D. $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

18. What is the purpose of transpilation?

- A. To convert a circuit into backend-compatible instructions
 - B. To measure all qubits automatically
 - C. To create a statevector
 - D. To define an observable
19. What does an X gate do to $|0\rangle$?
- A. Creates $|+\rangle$
 - B. Maps it to $|1\rangle$
 - C. Measures the qubit
 - D. Adds a global phase only

Answer Key

1. S3-001: B
2. S4-003: A
3. S6-002: B
4. S1-002: C
5. S5-002: A
6. S7-001: A
7. S8-002: A
8. S7-002: A
9. S4-002: B
10. S6-001: B
11. S8-001: A
12. S2-001: B
13. S5-001: B
14. S1-003: C
15. S2-002: C
16. S3-004: C
17. S3-002: C
18. S4-001: A
19. S1-001: B